

Aldrick Victor Tadeo Arellano

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Robotics and Digital Systems Engineering undergraduate focused on autonomous robotic systems, sim-to-real development, perception, SLAM, and deployable embodied AI. Seeking graduate study opportunities to advance research in robotic autonomy, digital-twin-based experimentation, and field-ready intelligent systems that transfer reliably from simulation to real-world environments.

Education

B.S. in Robotics and Digital Systems Engineering
Monterrey Institute of Technology and Higher Education (Tecnológico de Monterrey)

- Academic Excellence Scholarship recipient for sustained high academic performance.
- Specialization in Cyberphysical Systems, with emphasis on embedded systems, robotics, control, and intelligent automation.

2022 — 2026
CGPA: 95/100

Research Interests

- Sim-to-real robotics, digital twins, and simulation-based experimentation for autonomous systems.
- SLAM, perception, sensor fusion, and autonomous navigation using LiDAR, RGB-D, and visual sensing.
- Embodied AI and field robotics for robust deployment in uncertain, real-world environments.
- Robotic manipulation, soft robotic end-effectors, and perception-guided agricultural robotics.

Research and Academic Projects

LiDAR-Based SLAM for Autonomous Navigation
Manchester Robotics, Tec de Monterrey

- Developing a ROS2-based LiDAR SLAM pipeline for autonomous navigation on the MCR2 Puzzlebot platform, with emphasis on real-time execution on embedded compute hardware.
- Implementing scan matching and map-generation workflows to improve localization robustness and mapping consistency.
- Integrated Gazebo-based simulation workflows with physical robot testing to support iterative sim-to-real evaluation.
- Designed experimental protocols for maze mapping, solving, and navigation in simulated and real-world environments.
- Structured the system for future integration with multi-sensor fusion, autonomy stacks, and comparative SLAM benchmarking.

Mar 2026 — Present

Delicate Ripe Fruit Harvesting Robot
IEEE RoboSoft 2026 - Manipulation Challenge Competition

- Collaborated in a four-member team to design and build a robotic harvesting system using a UR5e manipulator, custom soft gripper, and in-hand RGB-D sensing for object detection and pose estimation.
- Jointly developed a soft gripper design inspired by raspberry morphology, incorporating granular jamming for stiffness modulation and dexterous, low-damage fruit interaction.
- Developed monolithic silicone manufacturing workflows using 3D-printed molds and primitive-based design strategies to improve fabrication consistency and gripper performance.

Aug 2025 — Apr 2026

Autonomous Harvesting Robot
Cyberphysical Systems Specialization Project

- Led development of a bio-inspired soft robotic gripper for strawberry harvesting and small-object picking within a cyberphysical systems capstone project.
- Designed and fabricated a three-finger silicone gripper using 3D-printed molds, iterative prototyping, and experimental manufacturing refinements.
- Developed embedded control software for gripper actuation, sensor integration, and manipulation testing on a UR3e robot arm.
- Built an AI vision pipeline for strawberry classification, optimized for deployment on both workstation and edge-compute devices.
- Implemented RGB-D-based object localization to estimate fruit pose for perception-guided autonomous picking.
- Integrated perception, manipulation, and gripper-control subsystems into a functional autonomous harvesting prototype.

Aug 2025 — Nov 2026

Computer Vision Based Autonomous Navigation
Manchester Robotics, Tec de Monterrey

- Developed ROS2-based autonomous navigation and control software on the MCR2 Puzzlebot platform for mobile robotics experimentation.
- Implemented traffic-sign and traffic-light recognition modules, improving environmental adaptability and perception reliability by 33%.

Feb 2025 — Aug 2025

- Combined visual perception with onboard sensing to reduce false positives and navigation decision errors by 15% during real-world tests.
- Redesigned the ROS2 node-topic architecture to improve modularity, data flow, and deployment robustness.
- Integrated Gazebo simulation, real-world testing protocols, and telemetry dashboards to support sim-to-real development and performance analysis.

Work and Technical Experience

Software Engineer InternNov 2025 — Present

GE Aerospace

- Designed a telemetry framework for internal CAD/CAE and PLM engineering tools, defining event models and PostgreSQL schemas for engineering workflow analysis.
- Integrated telemetry into Siemens NX-centered toolchains through C#/.NET services, REST interfaces, and NX Open extensions.
- Built Grafana dashboards to analyze adoption patterns, bottlenecks, and reliability trends across engineering software systems.

Telemetry Development Division Lead and FounderJun 2024 — Oct 2025

Electrum Performance Racing Team

- Founded and led the telemetry division, building a real-time data acquisition stack for high-stress electric vehicle environments.
- Developed embedded telemetry and serial communication solutions that reduced data-transmission failures by 15%.
- Engineered firmware and data workflows across multiple vehicle subsystems while coordinating technical collaboration, documentation, and cost optimization.

Honors, Leadership, and Service

600+ Hours of Community Service Award, Dept. Formación Social - Tecnológico de MonterreyApr 2026

- Recognized for accumulating over 600 hours of community service throughout my undergraduate program.

IEEE RoboSoft 2026 Participant, IEEE Robotics and Automation SocietyApr 2026

- Competitor in the Manipulation Challenge at the 9th IEEE International Conference on Soft Robotics hosted in Kanazawa, Japan. Member of the first team from Mexico to participate; contributed to soft robotics design, perception, and system integration.

STEM Workshops, Clinica KoiApr 2025 — Jun 2025

- Delivered hands-on robotics and engineering workshops for children and teenagers in vulnerable situations.
- Designed outreach activities using computer vision, gesture recognition, and interactive robotics to make STEM concepts accessible and motivating.

Expoingenierías 2025 Finalist, Tecnológico de MonterreyJul 2025

- Finalist in the engineering expo at Tecnológico de Monterrey, presenting a ROS2-based autonomous navigation system for mobile robotics applications.

Academic Excellence Scholarship, Tecnológico de Monterrey2022 — 2026

- Merit-based scholarship supporting undergraduate studies in recognition of academic performance and potential.

Technical Skills

Programming: Python, C/C++, C#, MATLAB, JavaScript, VHDL, Typst
Robotics: ROS2, SLAM, Gazebo, RViz, Nav2D, Sensor Integration, Robot Kinematics, Robot Control
AI and Perception: Machine Learning, Deep Learning, Computer Vision, Object Detection, Pose Estimation
Embedded and Hardware: ESP32, STM32, Arduino, Raspberry Pi, KiCAD, Circuit Design, FPGAs
Software and Tools: Linux, Docker/Podman, PostgreSQL, Grafana, Git/GitHub, RoboDK
CAD/CAE/SIM: Autodesk Fusion, SolidWorks, Siemens NX, AutoCAD, SimScale, Ansys Static Structural, Finite Element Analysis (FEA), Hyperelastic Material Modeling
Libraries and Frameworks: OpenCV, TensorFlow, PyTorch, NumPy, Ultralytics

Languages

English, Native ProficiencyIELTS: 8.5 (C2) 05/2026

Spanish, Native Proficiency

Japanese, Elementary Proficiency

Certifications

Fundamentals of Deep Learning, NVIDIACredential ID: 1tO0Ys3ITkGJkXM3sgBKrQ

Generative AI with Diffusion Models, NVIDIACredential ID: TauXuWfURMOBYNutOVkopw

OpenCV Bootcamp, OpenCV UniversityCredential ID: 65a25e083f50497dba5f5538026087

UR e-Series Tracks (Core, Pro, Application), Universal Robots